Mohamed Ait-Hocine

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Education

BSc Computer Science, Queen Mary University of London (1st Class with Honours) 2022 - 2025

Modules Include: Professional and Research Practice, Database Systems, Data Mining, Neural Networks and Deep Learning

- Researched the impact of traffic patterns on air quality in London, focusing on the effects of the Ultra Low Emission Zone.
- Modelled and visualised air quality data using the Python libraries Pandas and MatPlotLib to identify trends and patterns.
- Wrote a 10,000 word Dissertation about 'London Air Quality: Investigating Traffic Patterns and ULEZ Impacts'
- Presented dissertation findings to faculty and peers, receiving positive feedback for clarity and depth of analysis.

Relevant Experience

MedTech Hackathon Nov. 2023

Microsoft, Carradale Futures *London, UK*

- Deconstructed a problem brief regarding Electronic Patient Records (EPRs), then assigned tasks to members of a 5-man group, optimizing team efficiency.
- Designed a concept of an 'EPR lite' solution using Figma to address fundamental user requirements.
- Collaborated with group members to conceptualise a working prototype in under 24 hours using PowerApps, showcasing proficiency in low-code development.
- Presented the prototype to judges, articulating the project's objectives and outlining the seamless implementation of the user requirements.

Projects

Blockudoku Puzzle Game | *Java, Swing*

- Architected the game using Object Oriented Programming principles in Java, creating modular classes for game pieces.
- Engineered the core game logic, developing algorithms for grid validation and automatic line clearing of completed lines to dynamically update game state.
- Implemented custom data structures to represent the 9x9 game grid and over 15 unique block shapes.
- Developed a comprehensive testing suite using JUnit, achieving 100% code coverage for critical game logic modules, including scoring and block placement algorithms.

House Exploring Game | *Java, Swing*

- Architected the game using Object Oriented Programming principles in Java, creating modular rooms which peers can implement into their house.
- Integrated peer-created rooms from a central Git repository, validating the project's interoperability and modular design with only minor code adjustments required.
- Designed an intuitive user interface (UI) with Java Swing, ensuring a consistent user experience (UX) across different rooms and interactions.

Weather App for Football Fans | *React, OpenWeatherMap API*

- Led a team of 4 in the development of a weather application using OpenWeatherMap API to fetch and display data for football fans.
- Utilised the agile methodology to manage the project, ensuring timely delivery via regular progress meetings.
- Prototyped designs using Figma then used React to build the front-end interface.
- Created documentation and a 10-minute video walkthrough to help users understand the app's features and functionality.

FDM Expenses App | *Vite.js, Flask, PostgreSQL*

- Collaborated with a team of 7 to develop a full-stack expense management application using a waterfall development methodology.
- Designed a wireframe and mockup on Figma to visualize the user interface and user experience before development.
- Led the front-end development using Vite.js, working closely with 4 peers, ensuring a responsive and user-centered interface.
- Presented to FDM stakeholders, showcasing the application's features and user benefits. Achieved #1 FDM Expenses project in the university cohort.

Similar Hobbies Social Media App | *Django, Vue*

- Engineered a full-stack social application, developing a Django REST framework backend for secure user authentication and a dynamic, component-based front-end with Vue.js.
- Designed the core feature, a recommendation engine, by developing a matching algorithm in Python to query the SQL database and identify users with overlapping interests.
- Developed an intuitive user interface (UI) with Vue.js, creating reusable components for user registration, profile management, and hobby input forms to ensure a consistent user experience (UX).
- Conducted user feedback sessions with peers, systematically gathering insights to iterate on the UI/UX and improve the recommendation algorithm's accuracy.

Technical Skills

Languages: Java, Python, SQL (PostgreSQL, MySQL), JavaScript, HTML/CSS
Frontend: React, Vue, Vite.js, Bootstrap, Swing
Backend: Django, Flask
Developer Tools: Git, Conda, Docker, VS Code, Visual Studio, IntelliJ, Jupyter Notebooks, Wireshark